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Key concepts: Growth Hormone

1. Growth hormone (GH) is synthesized in somatotrophs in the anterior pituitary and is released in response to Growth Hormone-Releasing Hormone (GHRH) from the hypothalamus.
2. Growth hormone is secreted in a pulsatile diurnal manner, with the greatest secretion during deep, slow wave sleep (between Stage III and IV) and lowest secretion during the day.
3. Secretion of GH is stimulated by arginine, fasting, exercise, stress, hormones ghrelin, estrogen and testosterone, and low serum glucose and fatty acids.
4. Secretion of GH is inhibited by increased serum glucose and fatty acids, obesity, aging, somatostatin and IGF-1.
5. Secretion of GH is constant throughout childhood but surges at puberty from stimulation by estrogen and testosterone. GH levels decline to a stable level post-puberty and diminish further with aging.
6. Somatostatin is secreted by the hypothalamus and inhibits GH secretion by the anterior pituitary.
7. GH secretion is diminished by negative feedback at multiple levels: Somatomedins (IGF-1) inhibit release of GH in the anterior pituitary directly, and also stimulate secretion of somatostatin by the hypothalamus which in turn inhibits the release of GH by the anterior pituitary. GH also inhibits release of GHRH.
8. GH has direct actions through binding to GH receptors (linked to the JAK/STAT signaling pathway) in adipose tissue, muscle and liver and at the growth plate in bone. GH has indirect actions through stimulating the production of IGF-1 (Somatomedin C) in the liver.
9. Sufficient levels of insulin lead to greater GH receptor expression and signaling in hepatocytes.
10. IGFs are transported in serum in a complex with IGF binding protein (IGF-BP) and acid-labile subunit. GH stimulates production of IGF-1, IGF-BP3 and acid labile subunit in the liver. IGF-BPs inhibit IGF action but greatly extend the half-life of IGF. IGF-BP proteases in tissues degrade the binding protein and activate IGF locally.
11. In tissues, IGF-1 binds to the IGF-1 receptor, which is a tyrosine kinase receptor.
12. The actions of GH are broadly to stimulate growth and protein production in the fed state, and in the fasted state to promote lipolysis (spare glucose consumption).
13. The growth-promoting effects of GH are largely, though not completely, mediated through IGF-1.
14. GH stimulates cartilage growth and linear and appositional (increased diameter) bone growth. After closure of the epiphyses, GH continues to stimulate appositional growth.
15. When amino acids and glucose are in ample supply, GH levels increase, leading to amino acid uptake and protein synthesis in muscle and IGF-1 production by the liver.
16. When glucose but not amino acids are available, GH secretion is inhibited, promoting caloric storage but not growth.
17. In the fasted state, GH decreases glucose uptake in muscle, adipose tissue and stimulates gluconeogenesis in the liver, maintaining blood glucose. GH also stimulates lipolysis in adipose tissue, mobilizing fatty acids and switching metabolism to fat as the primary energy source.
18. GH antagonizes the action of insulin at the post-receptor level, and therefore decreases insulin sensitivity and is considered a diabetogenic hormone.
19. Excess GH can cause diabetes mellitus as well as gigantism before puberty and acromegaly post-puberty. GH deficiency can cause hypoglycemia as well as growth deficiencies in childhood. Laron syndrome originates from a mutation in the GH receptor that decreases sensitivity to GH.